

SPECIAL REPORT



Can large language models help in the assessment of people with pain?

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ABSTRACT

Chronic pain (CP) affects an important proportion of the global population, imposing a significant socio-economic burden and serving as a leading cause of disability. While the biopsychosocial approach is the gold standard for treatment, many patients fail to receive adequate care due to healthcare resource constraints. Effective management requires a comprehensive assessment. However, traditional questionnaires often miss the subjective and multidimensional depth of the pain experience. Differently, written narratives (WNs) empower patients to describe their pain in their own words. Nevertheless, qualitative analysis, such as WN, is often time-consuming for clinicians.

This paper explores how Large Language Models (LLMs) can facilitate clinicians in analyzing WNs. Evidence from the AINarratives project and other emerging studies indicates that LLMs can evaluate pain narratives with accuracy and utility comparable to human experts, identifying biopsychosocial themes that correlate with standardized clinical scores.

We conclude that LLMs should serve as clinical companions rather than substitutes, streamlining qualitative data processing to enhance the clinician-patient interaction and facilitate personalized care.

PLAIN LANGUAGE SUMMARY

Chronic pain affects millions of individuals, leading to a diminished and challenging quality of life. To enhance patients' well-being, healthcare professionals must understand how each person subjectively experiences pain. They typically rely on standardized measures; however, these tools often fail to capture the personal and multifaceted nature of living with pain. In recent years, an alternative that has been gaining ground is that of asking patients to write down their own stories about their pain. This lets people explain what truly matters to them using their own words. However, reading and analyzing these personal stories can be demanding and time-consuming for healthcare professionals. This article proposes a potential approach for using advanced Artificial Intelligence (AI) systems to assist healthcare professionals in analyzing patients' personal stories. Research supports that AI can quickly read patients' written stories and accurately pick out the most important details about how pain affects them from a biopsychosocial perspective. Most importantly, AI is meant to be a helpful tool, not a replacement for health care professionals. By quickly analyzing long texts, AI saves humans' time. This gives healthcare professionals more time to focus on patients, leading to better and more personal care.

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1. The impact of chronic pain

Chronic pain (CP) represents a major global health challenge, with studies reporting prevalences that can be higher than 30% when including low and middle-income countries [1], making it one of the most prevalent chronic conditions. It is associated with significant restrictions in daily activities, reducing functionality and quality of life, and impacting familial, professional, and social domains. According to the Global Burden of Disease Studies, pain and pain-related conditions are the leading causes of disability and disease burden worldwide [2]. Chronic pain is also often linked to mental health issues, including mood and anxiety disorders [3]. This issue significantly contributes to the demand for medical services, imposing a noteworthy economic burden on both individuals experiencing the pain and society at large. Altogether, the situation generates a very significant economic impact. For example, a recent study reported

a cost of 722.8 billion \$ for the management of CP in the United States, including care costs and loss of productivity [4].

With these figures, an adequate management of chronic pain is an urgent need that has been recognized as a human right in various international documents and should be a priority in any healthcare system [5]. In this context, substantial evidence accumulated over several decades supports the effectiveness of a multidisciplinary biopsychosocial approach in improving quality of life among individuals with chronic pain [6]. Besides that, a significant number of people are not receiving the care they need. For example, Breivik et al. [3] found that only 2% of persons with CP were being treated by a pain specialist, and 40% of persons with CP were not satisfied with the treatment they received. Moreover, CP patients often try multiple treatment options in search of relief, yet many of these approaches prove unsatisfactory [7]. This is partly due to the current inadequacy of

Article highlights

- The traditional healthcare systems often fail to provide the necessary multidisciplinary care due to resource and time constraints.
- An adequate assessment is fundamental to understanding the situation of the person with chronic pain (CP) and to designing the best intervention strategy for them.
- While questionnaires are standard tools used to do the assessment of persons with CP, qualitative methodologies such as written narratives (WNs) have the potential to capture the subjective, complex, and multidimensional nature of pain more effectively than questionnaires, empowering patients to share personal insights that standardized tools might miss.
- Preliminary studies show that large language models (LLMs) can be able to evaluate WNs with accuracy comparable to human experts.
- LLMs can provide accurate, useful, and usable assessments for healthcare professionals while helping patients better articulate their pain experiences.
- LLMs are envisioned as tools to augment, rather than replace, clinicians, streamlining qualitative data analysis while human professionals maintain essential supervision and patient interaction.

health systems in terms of physical facilities, materials, and specialized multidisciplinary teams to meet the high demand of individuals suffering from CP [8]. In any case, this situation increases the suffering of persons with CP due to feelings of frustration [9]. In this context, Information and Communication Technologies (ICTs) can help bridge the gap by providing timely and accessible resources currently unavailable through traditional healthcare systems [10]. Within this framework, this perspective paper examines how LLMs can support a key aspect of chronic pain management: pain assessment through written narratives (WN).

2. Importance of an adequate assessment of the pain experience

Conducting a comprehensive assessment of the pain experience is crucial to providing adequate care for persons with CP. This assessment process allows healthcare professionals to understand the particular pain experience of each subject and determine the most appropriate treatment strategies tailored to their needs. As emphasized by the International Association for the Study of Pain [11], pain is inherently a personal experience, reinforcing the need for individualized assessments and treatment plans. An adequate assessment is also necessary to understand the effects and efficacy of the interventions applied to persons with CP. Along these lines, the IMMPACT initiative, led by prominent researchers in the field, was one of the biggest efforts in the field to develop consensus recommendations to improve the design, execution, and interpretation of clinical trials for pain treatments. More specifically, the researchers identify core outcome domains that should be assessed, as specific measures for those outcomes [12]. Finally, an adequate assessment is essential for monitoring pain progression, identifying related variables, and implementing targeted strategies to help individuals manage these factors effectively. At the end, the assessment process

is like the compass that guides the entire process of helping a person with pain [13].

2.1. The use of questionnaires, qualitative approaches, and written narratives in the pain field

Questionnaires are widely used self-report measures to efficiently gather information about individuals, particularly for individual non-directly observable phenomena, such as the pain experience [12]. Nevertheless, while these tools supply valuable and essential insights into the issue, qualitative methodologies have been claimed to be more powerful in understanding the subjective nature of pain [14]. These offer the potential to reveal information about the private, subjective, intricate, and multidimensional aspects of pain that standardized questionnaires may not capture [15]. Moreover, patients can approach their pain experiences through their own words. Accordingly, they are gaining attention in the healthcare sector, which is increasingly recognizing the significance of narrative methodology as a specific way to access qualitative information [16].

According to Charon (cited in Carr et al. [17]), a narrative is “a history that has a teller, a listener, a language, characters, a plot, and time. Shopping lists and mathematical formulae are not narratives; novels, scripture, jokes, and newspaper articles are.” Narratives enable researchers to explore personal stories, allowing individuals to highlight the aspects they find most significant in their own words. Narratives have been employed by various scholars in the CP field. For instance, Noel et al. [18] analyzed narratives from interviews with 46 parents of young patients with CP. These interviews allowed Noel et al. [18] to identify two different clusters of narratives (based on content and affect codes): distress-focused and resilience narratives. The distress cluster was related to higher scores of helplessness, although both clusters did not differ in pain-related functioning. Similarly, Meldrum et al. [19] examined semi-structured interviews collecting narratives from 53 children experiencing CP at baseline and 6–12 months post-clinic intake. The authors identified five common themes related to isolation, changed self-perception, activity limitations, concerns about barriers to future goals, and lack of medical validation. Furthermore, a sixth theme was identified in relation to therapeutic change (from the interview conducted between 6 and 12 months after following the intervention) for some of the children.

The use of narrative methodology has extended to WN as well. For example, WNs have been used, among others, by McGowan et al. [20] in 32 women with chronic pelvic pain. Together with other relevant findings, it was found that most participants have an illness trajectory often characterized by problems in receiving adequate management and treatment. Relevant themes emerging from narratives included patients’ experiences of disengagement, the perceived need to obtain objective evidence of their condition, and the search for validation and recognition. Dysvik et al. [21] used WN (together with images) in 5 women with CP six years after completing a pain management intervention. Narratives about the experience showed grief related to the loss of usual lives and revealed loneliness and despair. Participants’ present situation

reflects adaptation and hope for the future, with pain management intervention acting as an important driver. In another study, Bergen et al. [22] used WN in 37 women with endometriosis. Their results identified three relevant themes for the participants related to the stigma and disruption in quality of life, the barriers to receiving adequate healthcare (including difficulties in the diagnosis process), and the need to rely on themselves and social support to cope with endometriosis. In a different investigation, Göthersson et al. [23] explore pain and fear related to negative experiences of needle procedures in 83 children/adolescents with diabetes using WNs. Results revealed that negative experiences have many facets, are generally hard, changing, and affect over time. Recently, a group of researchers used WN successfully in 39 children with functional abdominal pain, and their parents [24], 92 persons with neck and/or back pain [25], and 46 persons with fibromyalgia [9]. Besides the specificities of each sample, results allow understanding patients' pain experience, identifying relevant pain triggers, coping strategies, treatments followed, participant trajectories, and difficulties encountered in receiving appropriate diagnosis and management. Together with prior literature, the results of Nieto et al. [24] support the feasibility of using WN in a wide range of pain populations.

In conclusion, empowering patients to freely articulate their experiences of pain and suffering from their own perspective, empirical research indicates that WNs allow researchers to understand patients' own histories, which are generally characterized by a perceived lack of support and understanding (often linked to challenges in achieving accurate diagnosis and adequate clinical management).

WNs offer several advantages over traditional oral interviews, as writing supports a clearer organization of thoughts and emotions, particularly in relation to complex experiences such as pain [26]. Patients are able to decide what to express and how to express it, thereby setting their own agenda, prioritizing the aspects most relevant to their perspective, and describing their experiences without the immediate presence or influence of a clinician [20]. Requesting a WN prior to an initial pain management consultation may provide clinicians with a more comprehensive understanding of each patient's situation and needs, thereby improving the efficiency of the visit [24]. In practice, patients can complete the task outside the clinical setting, helping to address the limited time typically available for pain assessments. Overall, this approach enables professionals and researchers to capture the complexity of individual experiences effectively, revealing relevant aspects that are often overlooked by structured instruments, such as self-report questionnaires, and optimizing the use of time prior to the first consultation [14,15].

3. Can LLMs help?

Besides the large potential of qualitative methodologies in general, and WNs in particular, more work in the area is needed to potentiate its use both in research and clinical contexts.

LLMs are a class of AI models capable of processing and generating human-like text. Their ability to seemingly understand and generate text in a contextually relevant and

coherent manner makes them highly versatile, with applications in text generation, summarization, classification, and assessment. Accordingly, LLMs are a powerful tool for analyzing textual inputs such as WNs. Importantly, they can also be applied to other forms of narrative data, including spoken interviews, provided that these are first transcribed into written text. In this regard, a wide range of Speech-to-Text (STT) technologies are now available, and such capabilities are increasingly integrated into modern LLM systems. It is worth noticing that in those cases where interview-based dimensions such as tone and prosody are considered relevant for patient assessment, the overall structure of the proposed approach remains unchanged when shifting from written to spoken input. In particular, the core conceptual framework, evaluation methodology, and ethical considerations are preserved. The main adaptation concerns the specific LLMs configuration, which may be extended to incorporate additional features for tone and prosody recognition.

In light of these considerations, we hypothesize that the use of LLMs may support clinicians in the assessment of patients' WN, while also alleviating the time constraints that can hinder the delivery of adequate pain evaluation [27].

The capabilities of LLMs have been perceived by healthcare professionals who, according to recent studies, are, in the majority, open to using them. For example, Egli et al. [28] found that 69.3% of Swedish physicians surveyed had used an LLM for general purposes at least once, and 32.8% were frequent users. When asked about the use of LLMs for medical purposes, 56.9% reported using them at least once, and 21.8% were frequent medical users.

3.1. Evidence from the literature

In line with our hypothesis, a recent study examined the capability of GPT-4 to analyze WNs from persons suffering from fibromyalgia [29]. Their findings indicated that GPT-4's evaluations were consistent with those of human experts, who, furthermore, deemed accurate and useful the scoring explanations generated by GPT-4. In addition, GPT-4 WNs' assessment showed positive correlations with scores obtained from standardized questionnaires. From these results stemmed the AINarratives project, which aims to enhance the assessment process for CP patients. AINarratives is envisioned as an LLM-powered platform with two target groups: patients and healthcare professionals. For patients, it will support pain narration, with AINarratives providing general guidelines for producing WNs. For professionals, it will support the assessment phase by providing scores for multiple pain dimensions (e.g., intensity and disability) as expressed by WNs, along with the explanation for those scores. Based on its use by 14 healthcare professionals [30], the results confirm the findings of Amidei et al. [29], the scores provided by the platform are rated as accurate and useful, and indicate that the platform was rated as usable according to the SUS [31].

Beyond these preliminary works directly focused on using LLMs to analyze WN of patients with CP, there are emerging related works that support the use of LLMs to assess pain by analyzing pieces of text in general. Walker et al. [32] analyzed Reddit narratives contained in 31,785 posts about Lupus Pain.

They fine-tune the LLaMA-3.2-3B-Instruct model to analyze the contents and provide structured summaries according to the tenets of the biopsychosocial model. The system was able to accurately identify topics related to the biological, psychological, and social dimensions of pain. Flaherty and Oliver [33] qualitatively analyze open responses from 82 patients with CP who had participated in an intervention program by using GPT. Specifically, the patients were asked about what was most helpful for them from the intervention they had followed, or whether they had learnt anything in particular that might be helpful to them in the future. GPT was able to identify seven different major themes relevant to evaluating the intervention program, providing rich information. Nunes et al. [34] used a set of NLP pipelines to process patients' personal descriptions of pain (which were transcribed from verbal interviews), and analyze to what extent these allow for predicting important variables related to the pain experience. Their findings suggest that LLMs, particularly larger models, were able to predict the outcomes; especially in predicting pain intensity, depression, positive affect, and anxiety. However, correlations were modest and not completely precise. Finally, it is worth noticing that LLMs have also been used fruitfully to analyze qualitative content in other health problems [35].

Additionally, other works have highlighted the ability of LLMs to operate with text related to clinical notes. For example, Vaid et al. [36] used locally running, privacy-preserving LLMs capable of following plain language instructions to extract characteristics of musculoskeletal pain (such as location and acuity) from a heterogeneous collection of unstructured clinical notes. The study used multiple patient notes, coded by 2 health care professionals, and found high precision of the system in classifying pain location and acuity. Patel et al. [37] compared traditional NLP models with GPT-4 and PaLM-2 for the extraction of patient-reported outcomes. Their findings, focused on persons with abdominal pain, showed that LLMs outperform traditional NLP approaches, achieving high accuracy with physician scores. Also in this case, the results highlight the potential benefits of employing LLMs in healthcare. Liu et al. [38] introduce a web-based system (AMTES) built on DeepSeek-V2.5 to support history-taking training in medical education. AMTES was evaluated by 31 medical students across three use cases (created by experts for the training simulation task), one of which focused on abdominal pain. Results indicate that AMTES effectively simulates patient interactions, providing high-quality dialogue and transparent evidence-based feedback. The authors concluded that AMTES holds significant potential as a history-taking training tool in medical education.

3.2. Ethical, technical, and methodological considerations for the use of LLMs in the pain field

The use of LLMs can rely on both commercial systems, such as Gemini, GPT, or Claude, and open-source models, such as Llama or DeepSeek. Although their performances may appear comparable, important ethical, technical, and economic differences must be considered. In particular, issues related to data governance, privacy, model decommissioning, reproducibility,

and long-term accessibility can emerge when relying on commercial solutions. Conversely, locally deployed open-source models may reduce some of these concerns but require advanced technical expertise and costly computational infrastructures.

Considering these trade-offs, we believe that commercial LLMs may be appropriate during exploratory research phases aimed at testing hypotheses related to pain management applications, provided that their limitations are clearly acknowledged. In this case, researchers are in charge of preserving the anonymity of data; for example, practical safeguards should include de-identification or pseudonymization before model processing, contractual restrictions on vendor data retention and secondary use, encryption, access control, audit logging, and retention limits. However, in later stages involving real medical deployment, open-source models hosted within secure local infrastructures should be preferred to ensure stronger control over privacy and governance, while complying with relevant regulatory frameworks. For example, in the United States, an LLM-based system may fall under FDA oversight as Software as a Medical Device (SaMD) if it is deployed with a medical intended use, such as informing diagnosis or treatment. In Europe, LLM-based systems used for medical purposes may also fall under the EU Medical Device Regulation and the EU AI Act, depending on intended use and deployment context. The European Commission describes high-risk AI systems, including AI-based software intended for medical purposes, as requiring risk mitigation, high-quality datasets, clear user information, and human oversight.

Regarding model development and evaluation, all stages should comply with FAIR (Findability, Accessibility, Interoperability, and Reuse) principles. Prompt-engineered systems should be systematically tested across multiple cases, with transparent reporting of prompt selection procedures. Reproducibility and output consistency must be evaluated through repeated testing on the same input. In fine-tuned systems, datasets should be carefully curated to minimize imbalance and bias. We further recommend integrating explainability mechanisms, such as requiring models to provide evidence-linked rationales for their assessments (still, these rationales should be treated as model outputs requiring validation rather than faithful explanations of the model's internal reasoning), and Retrieval-Augmented Generation (RAG) [39] frameworks may be adopted when appropriate to ground outputs in auditable source material and potentially reduce hallucinations, though they still require independent validation. Finally, both experts and patients should be involved throughout development and evaluation, and all assessments should be repeated whenever a model is replaced or updated.

4. Conclusions

LLMs provide a promising opportunity to enhance the understanding and assessment of CP, particularly by supporting qualitative evaluations of patients' pain experiences. By analyzing WN descriptions and linguistic patterns, these models

can contribute additional insights that may complement traditional assessment methods. While the existing literature discussed in this paper lends support to the feasibility and potential value of this approach, it also makes clear that further investigation is required, and ethical and methodological considerations must be taken into account. More comprehensive research is necessary to rigorously evaluate the reliability, consistency, reproducibility, and clinical usefulness of LLMs when they are applied to tasks related to pain assessment.

This need for caution is further underscored by growing evidence that LLMs are not neutral or bias-free systems. Prior studies (e.g., Gallegos et al. [40]) have shown that biases embedded in training data or model design can influence generated outputs, potentially affecting interpretations of patient WN. In a clinical context, such distortions could have meaningful consequences if they are not carefully identified and managed. At the same time, individuals living with CP frequently report concerns about the increasing role of technology in healthcare, particularly fears that digital tools may reduce opportunities for direct, empathetic contact with healthcare professionals [7]. These concerns highlight the importance of building trust in human interaction in health-related contexts. Taken together, these issues make it crucial to ensure that LLM-based assessments are robust, transparent, and dependable. Particular attention must be paid to minimizing risks such as hallucinations or inaccurate interpretations in model-generated outputs, especially when these tools are used to support clinically relevant analyses of patient-reported experiences.

Ultimately, while LLMs can offer valuable support for CP assessment through WN analysis, they should be understood as supportive tools rather than replacements. Healthcare professionals remain indispensable for oversight, interpretation, and the provision of compassionate, patient-centered care.

5. Future perspective

LLMs are gaining popularity and an important role in our everyday lives. We hypothesize that they will also be adopted extensively in the healthcare system. In the pain field, LLMs can be useful to facilitate the assessment of pain from a qualitative point of view. By analyzing WNs from persons with CP, which capture patients' subjective perspectives, LLMs can accelerate clinicians' assessment processes. Such assessments can complement (or, in some cases, match) the value of traditional self-report questionnaires by highlighting what matters most to patients and supporting more tailored interventions.

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